DRUGS · ALLERGY, RHINITIS AND BIOLOGICS

Mepolizumab and anti-IL-5 biologics

Mepolizumab, reslizumab and benralizumab target interleukin-5, the cytokine that drives eosinophil maturation and survival. They are used in severe eosinophilic asthma, severe chronic rhinosinusitis with nasal polyps, and eosinophilic granulomatosis with polyangiitis (EGPA), a vasculitis that ENT surgeons may be the first to suspect in a patient with polyps, asthma and eosinophilia.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

ANTI-IL-5 BIOLOGICS

	SUMMARY
Agents	Mepolizumab and reslizumab (bind IL-5); benralizumab (binds the IL-5 receptor alpha and depletes eosinophils by antibody-dependent cytotoxicity)
How they work	Reduce eosinophil production, survival and tissue infiltration
ENT uses	Severe CRS with nasal polyps (mepolizumab is licensed in many countries); severe eosinophilic asthma; EGPA (mepolizumab 300 mg)
Doses	Mepolizumab 100 mg subcutaneously every 4 weeks (300 mg every 4 weeks in EGPA); benralizumab 30 mg every 4 weeks for 3 doses, then every 8 weeks; reslizumab intravenously by weight every 4 weeks
Pregnancy	Limited data; use only if needed

2 CHAPTER TWO Using them in ENT

Indications.

2.1 Polyps and asthma

For patients meeting the EPOS biologic criteria, especially those with blood eosinophilia and severe eosinophilic asthma, mepolizumab reduces polyp size and steroid use and the need for surgery. Response is assessed at 6 months and annually.

2.2 Recognising EGPA

Adult-onset asthma, nasal polyps, eosinophilia above 10%, and then neuropathy, purpura, cardiac or renal involvement suggest EGPA (formerly Churg-Strauss syndrome); about 40% are ANCA (MPO) positive. Refer to rheumatology; mepolizumab is used for relapsing or refractory disease.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

- Injection-site reactions, headache, back pain; hypersensitivity rarely.
- **Herpes zoster** has been reported: consider vaccination before starting.
- Helminth infection: treat before starting.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

> **WHAT THE EXAMINER IS LOOKING FOR**
>
> - The difference between IL-5 and IL-5 receptor antibodies.

- Features of **EGPA** and why ENT surgeons should spot it.

4.2 Common pitfalls

COMMON PITFALLS

- Missing EGPA in a patient with polyps, asthma and marked eosinophilia.

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